

Roll No.....

MID SEMESTER EXAMINATION MARCH- 2025

Name of the Program: B.Tech.

Name of the course: Design Of Machine Elements-II

Time: 90 Minutes

Semester: Sixth

Paper Code: TME602

Maximum Marks: 50

Note: -

- (i) Answer all questions by choosing any one of the subsections
- (ii) Each question carries 10 marks

Q1.	10 marks	
(a)	Classify different types of springs and briefly describe their applications.	CO1
(b)	Explain the construction and working principle of a leaf spring. Derive the expression for bending stress in a leaf spring and discuss the conditions for achieving equalized stress distribution.	
Q2.	10 marks	
(a)	Explain the important terminologies used in helical springs. Also, derive the expression for the curvature effect in helical springs.	CO1
(b)	A helical compression spring is subjected to a fluctuating load ranging from 400 N to 1000 N. The spring has a spring index (C) of 6 and a factor of safety (Fs) of 1.25. The yield strength in shear (τ_{ys}) is 770 MPa, and the endurance strength in shear (τ_e) is 350 MPa. Given that the maximum compression of the spring under peak load is 30 mm and the modulus of rigidity (G) is 80 kN/mm ² , determine the following design parameters: (i) Wire diameter (d) (ii) Mean coil diameter (D) (iii) Outer and inner coil diameters (D_o and D_i) (iv) Number of active coils (n)	
Q3.	10 marks	
(a)	A helical spring is subjected to fluctuating loads. Derive the governing equation for designing a spring for fatigue loading, considering the endurance limit and safety factor. Explain how fatigue failure is avoided in spring design.	CO1/ CO2
(b)	A cross belt drive system consists of two pulleys with diameters of 450 mm and 200 mm, mounted on parallel shafts that are 1.95 m apart. Determine the following: (i) Total length of the belt (ii) Angle of lap on both pulleys (iii) Power transmitted when the larger pulley rotates at 2000 rpm (iv) Tension ratio, assuming a coefficient of friction $\mu = 0.25$	
Q4.	10 marks	
(a)	A helical tension spring is used in a spring balance to measure weights. One end of the spring is fixed to a rigid support, while the free end carries the weight to be measured. The maximum weight the spring balance should measure is 1800 N, and the length of the scale should be approximately 120 mm. The spring index is given as 7. The spring is made of oil-hardened and tempered steel wire with an ultimate tensile	CO1 /CO2

	strength of 1400 N/mm^2 and a modulus of rigidity of $82,500 \text{ N/mm}^2$. The permissible shear stress in the spring wire is 50% of the ultimate tensile strength. Design the spring and determine the following parameters: (i) Wire diameter (ii) Mean coil diameter (iii) Number of active coils (iv) Required spring rate (v) Actual spring rate	
(b)	What is slip and creep in belt drives? Explain their causes, effects, and how they can be minimized in flat belt systems.	
Q5.	10 marks	
(a)	Derive the expression for centrifugal tension in a belt. How does centrifugal tension affect the power transmission capacity of the belt?	C01/ CO2
(b)	Derive the equation for the moment and torque in a helical torsion spring. Explain how it differs from a conventional helical compression spring.	